

Esercizi:

- V , $\dim V = 3$ $\mathbb{R}$ $\mathcal{B} = (e_1, e_2, e_3)$ base di V

$$T: V \rightarrow V \quad A = M_{\mathcal{B}}(T) = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$|A - \lambda I_3| = \begin{vmatrix} 1-\lambda & 0 & 1 \\ 0 & 1-\lambda & 1 \\ 0 & 0 & -\lambda \end{vmatrix} = (1-\lambda)^2(-\lambda) = 0 \Leftrightarrow \lambda_1 = 1 \quad 0 \quad \lambda_2 = 0$$

$$m_A(1) = 2, \quad m_A(0) = 1$$

$$U_0: A \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{cases} x_1 + x_3 = 0 \\ x_2 + x_3 = 0 \end{cases} \quad \begin{cases} x_1 = -x_3 \\ x_2 = -x_3 \end{cases} \quad \mathcal{S}_0 = \{(-x_3, -x_3, x_3) \mid x_3 \in \mathbb{R}\} = \mathcal{L}((-1, -1, 1)) = \phi_{\mathcal{B}}(U_0)$$

$$(-x_3, -x_3, x_3) = x_3(-1, -1, 1)$$

$$U_0 = \mathcal{L}(-e_1 - e_2 + e_3) = \text{Ker}(T)$$

$$U_1: \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{cases} x_3 = 0 \end{cases} \quad \mathcal{S}_0 = \{(x_1, x_2, 0) \mid x_1, x_2 \in \mathbb{R}\} = \mathcal{L}((1, 0, 0), (0, 1, 0)) = \phi_{\mathcal{B}}(U_1)$$

$$(x_1, x_2, 0) = x_1(1, 0, 0) + x_2(0, 1, 0)$$

$$U_1 = \mathcal{L}(e_1, e_2)$$

autovettori della matrice
autovettori dell'endomorfismo

$$\overline{\mathcal{B}} = \{-e_1 - e_2 + e_3, e_1, e_2\} \text{ base spettrale}$$

Una matrice che diagonalizza

$$Q = \begin{pmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} = M_{\overline{\mathcal{B}}\mathcal{B}}(\text{id}_V)$$

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = Q^{-1} A Q$$

T e A diagonalizzabili.

- $T: V \rightarrow V$ $\mathcal{B} = (e_1, e_2, e_3)$ $\mathbb{R}$

$$A = M_{\mathcal{B}}(T) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

$$? \mu: \phi_{\mathcal{B}}(\mu) = (0, 1, 0) ?$$

$$\mu = 0 \cdot e_1 + 1 \cdot e_2 + 0 \cdot e_3$$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & -\lambda & 1 \\ 1 & 0 & -\lambda \end{vmatrix} = \lambda^2(1-\lambda) = 0 \Leftrightarrow \lambda_1 = 0 \quad 0 \quad \lambda_2 = 1$$

$$m_A(0) = 2 \quad m_A(1) = 1$$

$$U_0: \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{cases} x_1 = 0 \\ x_3 = 0 \end{cases} \quad \mathcal{S}_0 = \{(0, x_2, 0) \mid x_2 \in \mathbb{R}\} = \mathcal{L}((0, 1, 0)) = \phi_{\mathcal{B}}(U_0) \Rightarrow U_0 = \text{Ker}(T) = \mathcal{L}(e_2)$$

$$U_1: \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{cases} x_1 = x_3 \\ x_2 = x_3 \end{cases} \quad \mathcal{S}_0 = \{(x_3, x_3, x_3) \mid x_3 \in \mathbb{R}\} = \phi_{\mathcal{B}}(U_1) = \mathcal{L}((1, 1, 1)) \Rightarrow U_1 = \mathcal{L}(e_1 + e_2 + e_3)$$

Non c'è base spettrale.

T e A non sono diagonalizzabili:

$$\dim(U_0 \oplus U_1) = \dim U_0 + \dim U_1 = 1 + 1 = 2 \neq 3$$

Non c'è bene ripartire.

T e A non sono diagonalizzabili: $\dim(U_0 \oplus U_1) = \dim U_0 + \dim U_1 = 1 + 1 = 2 \neq 3$

• $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ $\mathcal{B} = ((1,0,0), (0,1,0), (0,0,1))$ $\mathbb{R}$

$$A = M_{\mathcal{B}}(T) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 1 & 1 & 0 \end{pmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & -1 \\ 1 & 1 & -\lambda \end{vmatrix} = (1-\lambda) \begin{vmatrix} 1-\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = (1-\lambda) [-\lambda(1-\lambda) + 1] =$$

$$= (1-\lambda) [-\lambda + \lambda^2 + 1] = 0 \iff \lambda_1 = 1 \text{ oppure } \lambda^2 - \lambda + 1 = 0$$

$$\text{ma } m_{\mathbb{C}}(1) = 1 \quad \lambda = \frac{1 \pm \sqrt{1-4}}{2} \in \mathbb{C}$$

$$U_1: \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 1 & 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \quad \begin{cases} x_3 = 0 \\ x_1 + x_2 = 0 \end{cases} \quad \mathcal{S}_0 = \{ (-x_2, x_2, 0) \mid x_2 \in \mathbb{R} \} =$$

$$= \mathcal{L}((-1, 1, 0)) = \phi_{\mathcal{B}}(U_1) = U_1$$

↗ è l'identità.

T e A non sono diagonalizzabili.

• $T: V \rightarrow V$ $\mathcal{B} = (\mu_1, \mu_2, \mu_3)$ $\mathbb{R}$

$$A = M_{\mathcal{B}}(T) = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{pmatrix}$$

$$|A - \lambda I_3| = \begin{vmatrix} 4-\lambda & 2 & 2 \\ 2 & 4-\lambda & 2 \\ 2 & 2 & 4-\lambda \end{vmatrix} = (4-\lambda) \begin{vmatrix} 4-\lambda & 2 \\ 2 & 4-\lambda \end{vmatrix} - 2 \begin{vmatrix} 2 & 2 \\ 2 & 4-\lambda \end{vmatrix} + 2 \begin{vmatrix} 2 & 4-\lambda \\ 2 & 2 \end{vmatrix} =$$

$$= (4-\lambda) [(4-\lambda)^2 - 4] - 2 [2(4-\lambda) - 4] + 2 [4 - 2(4-\lambda)] =$$

$$= (4-\lambda) [16 - 8\lambda + \lambda^2 - 4] - 2 [8 - 2\lambda - 4] + 2 [4 - 8 + 2\lambda] =$$

$$= (4-\lambda) [12 - 8\lambda + \lambda^2] - 2 [4 - 2\lambda] + 2 [-4 + 2\lambda] =$$

$$= 48 - 32\lambda + 4\lambda^2 - 12\lambda + 8\lambda^2 - \lambda^3 - 8 + 4\lambda - 8 + 4\lambda =$$

$$= -\lambda^3 + 12\lambda^2 - 36\lambda + 32 = 0$$

$$1 \quad -1 + 12 - 36 + 32 \neq 0$$

$$-1 \quad 1 + 12 + 36 + 32 \neq 0$$

$$2 \quad -8 + 48 - 72 + 32 = 0$$

Teorema di Ruffini
 $\Rightarrow (\lambda - 2)$ divide il polinomio caratteristico

$$\begin{array}{r|l} -\lambda^3 + 12\lambda^2 - 36\lambda + 32 & \lambda - 2 \\ +\lambda^3 - 2\lambda^2 & -\lambda^2 + 10\lambda - 16 \\ \hline 10\lambda^2 - 36\lambda + 32 & \\ -10\lambda^2 + 20\lambda & \\ \hline -16\lambda + 32 & \\ +16\lambda - 32 & \\ \hline 0 & \end{array}$$

$$|A - \lambda I| = (\lambda - 2)(-\lambda^2 + 10\lambda - 16) =$$

$$= (2 - \lambda)(\lambda^2 - 10\lambda + 16) =$$

$$\lambda = 5 \pm \sqrt{25 - 16} = 5 \pm \sqrt{9} = 5 \pm 3 =$$

$$= \begin{cases} \lambda_1 = 8 \\ \lambda_2 = 2 \end{cases}$$

$$= (2 - \lambda)(\lambda - 2)(\lambda - 8) = -(\lambda - 2)^2(\lambda - 8)$$

$$m_{\mathbb{C}}(2) = 2, \quad m_{\mathbb{C}}(8) = 1$$

$$U_2: \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \quad \begin{cases} x_1 + x_2 + x_3 = 0 \end{cases} \quad \mathcal{S}_0 = \left\{ (-x_2 - x_3, x_2, x_3) \mid \begin{smallmatrix} x_2 \\ x_3 \end{smallmatrix} \in \mathbb{R} \right\} =$$

$$= \mathcal{L}((-1, 1, 0), (-1, 0, 1)) = \phi_{\mathcal{B}}(U_2)$$

$$U_2: \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \quad \left\{ \begin{array}{l} x_1 + x_2 + x_3 = 0 \\ x_2 = x_3 \end{array} \right. \quad \mathcal{B}_0 = \{(-x_2 - x_3, x_2, x_3) \mid x_2, x_3 \in \mathbb{R}\} = \mathcal{L}((-1, 1, 0), (-1, 0, 1)) = \phi_{\mathcal{B}}(U_2)$$

$$U_2 = \mathcal{L}(-u_1 + u_2, -u_1 + u_3)$$

$$U_8: \begin{pmatrix} -4 & 2 & 2 \\ 2 & -4 & 2 \\ 2 & 2 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$0 = -6x_2 + 6x_3$$

$$\underline{Q}^1 \rightarrow \underline{Q}^2 \quad \begin{pmatrix} 2 & -4 & 2 \\ -4 & 2 & 2 \\ 2 & 2 & -4 \end{pmatrix} \quad \begin{array}{l} \underline{Q}^2 \rightarrow \underline{Q}^2 + 2\underline{Q}^1 \\ \underline{Q}^3 \rightarrow \underline{Q}^3 - \underline{Q}^1 \end{array} \quad \begin{pmatrix} 2 & -4 & 2 \\ 0 & -6 & 6 \\ 0 & 6 & -6 \end{pmatrix} \quad \underline{Q}^3 \rightarrow \underline{Q}^3 - \underline{Q}^2 \quad \begin{pmatrix} 2 & -4 & 2 \\ 0 & -6 & 6 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\left\{ \begin{array}{l} x_1 - 2x_2 + x_3 = 0 \\ x_2 = x_3 \end{array} \right. \quad \left\{ \begin{array}{l} x_1 - x_2 = 0 \\ x_2 = x_3 \end{array} \right. \quad \left\{ \begin{array}{l} x_1 = x_2 \\ x_2 = x_3 \end{array} \right. \quad \mathcal{B}_0 = \{(x_2, x_2, x_2) \mid x_2 \in \mathbb{R}\} = \mathcal{L}((1, 1, 1)) = \phi_{\mathcal{B}}(U_8)$$

$$\Rightarrow U_8 = \mathcal{L}(u_1 + u_2 + u_3)$$

$$\overline{\mathcal{B}} = \{-u_1 + u_2, -u_1 + u_3, u_1 + u_2 + u_3\} \quad \text{basis spectrale}$$

$$Q = \begin{pmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix} \quad Q^{-1}AQ = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{pmatrix} = M_{\overline{\mathcal{B}}}(\tau) = \overline{A}$$

$$\text{Ricci's lemma: } u \in V, \quad u \equiv_{\overline{\mathcal{B}}} (\bar{x}_1, \dots, \bar{x}_m) \quad u \equiv_{\mathcal{B}} (x_1, \dots, x_m) \\ \tau(u) \equiv_{\overline{\mathcal{B}}} (\bar{y}_1, \dots, \bar{y}_m) \quad \tau(u) \equiv_{\mathcal{B}} (y_1, \dots, y_m)$$

$$\begin{aligned} A \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} &= \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} & \overline{A} \begin{pmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_m \end{pmatrix} &= \begin{pmatrix} \bar{y}_1 \\ \vdots \\ \bar{y}_m \end{pmatrix} \\ Q \begin{pmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_m \end{pmatrix} &= \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} & A Q \begin{pmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_m \end{pmatrix} &= Q \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} \Rightarrow \overline{A} = Q^{-1} A Q. \\ Q \begin{pmatrix} \bar{y}_1 \\ \vdots \\ \bar{y}_m \end{pmatrix} &= \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} & \Rightarrow Q^{-1} A Q \begin{pmatrix} \bar{x}_1 \\ \vdots \\ \bar{x}_m \end{pmatrix} &= \begin{pmatrix} \bar{y}_1 \\ \vdots \\ \bar{y}_m \end{pmatrix} \end{aligned}$$

$$E = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 8 & 2 & 2 \end{pmatrix} \quad E^{-1}AE = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$\bullet A_{\kappa} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & \kappa \\ 0 & \kappa & 1 \end{pmatrix} \quad \kappa \in \mathbb{R} \quad \text{Per quel valore di } \kappa \text{ la matrice } A_{\kappa} \text{ è diagonalizzabile?}$$

$$|A_{\kappa} - \lambda I_3| = \begin{vmatrix} 1-\lambda & 1 & 0 \\ 0 & 1-\lambda & \kappa \\ 0 & \kappa & 1-\lambda \end{vmatrix} = (1-\lambda) [(1-\lambda)^2 - \kappa^2] = (1-\lambda) [1 - 2\lambda + \lambda^2 - \kappa^2] = 0 \Leftrightarrow$$

$$\Leftrightarrow \lambda = 1 \quad \text{opp.} \quad \lambda^2 - 2\lambda + 1 - \kappa^2 = 0$$

$$\lambda = 1 \pm \sqrt{1 - 1 + \kappa^2} = 1 \pm |\kappa| = 1 \pm \kappa$$

$$\lambda_1 = 1, \quad \lambda_2 = 1 - \kappa, \quad \lambda_3 = 1 + \kappa$$

Se $\kappa=0$, $\lambda=1$ $\text{ma}(1)=3$

$$U_1: \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \quad \dim U_1 = 2 \neq 3 \quad \text{non e' diag.}$$

$$1-\kappa = 1+\kappa \Leftrightarrow 1-2\kappa = 1 \Leftrightarrow \kappa=0$$

Se $\kappa \neq 0$, abbiamo 3 autovalori due a due distinti
diagonalizzabile.

• $\dim E = 3$, $R = (O, B)$ ortonormale

$$r: \begin{cases} x+y-z+2=0 \\ 2x-y+z=0 \end{cases}$$

(i) Determinare il piano π ortogonale a r e passante per il punto $A(1,0,1)$.

$$\vec{r}: \begin{cases} x+y-z=0 \\ 2x-y+z=0 \end{cases} \quad \begin{pmatrix} 1 & 1 & -1 \\ 2 & -1 & 1 \end{pmatrix} \xrightarrow{R_2 - 2R_1} \begin{pmatrix} 1 & 1 & -1 \\ 0 & -3 & 3 \end{pmatrix} \begin{cases} x_1 = 0 \\ x_2 = x_3 \end{cases}$$

$$\vec{r} = \mathcal{L}(u(0,1,1))$$

$$\pi: x_2 + x_3 + d = 0 \quad A \in \pi \Rightarrow 0 + 1 + d = 0 \Rightarrow d = -1$$

$$\pi: x_2 + x_3 - 1 = 0 \quad w = u(0,1,1)$$

(ii) Determinare due differenti piani ortogonali a π e passanti per $B(1,1,0)$

$$\mathcal{H}: ax + by + cz + d = 0 \quad \mathcal{H} \perp \pi \Leftrightarrow (a,b,c) \cdot (0,1,1) = 0 \Leftrightarrow b+c=0 \Leftrightarrow b=-c$$

$$(1,0,0) \quad (0,1,-1)$$

$$\mathcal{H}': x_1 + \kappa' = 0 \quad B \in \mathcal{H}' \Rightarrow 1 + \kappa' = 0 \Rightarrow \kappa' = -1 \Rightarrow \mathcal{H}': x_1 - 1 = 0$$

$$\mathcal{H}'': x_2 - x_3 + \kappa'' = 0 \quad B \in \mathcal{H}'' \Rightarrow 1 - 0 + \kappa'' = 0 \Rightarrow \kappa'' = -1 \Rightarrow \mathcal{H}'': x_2 - x_3 - 1 = 0$$

(iii) Determinare il piano parallelo a r , ortogonale a π' : $-2x+y-z+1=0$
e passante per $C(1,2,-1)$

(iv) Data la retta $S: \begin{cases} x = t \\ y = 1-t \\ z = 3+t \end{cases}$ dire se S e π sono sghembe e se sono ortogonali.